Lighthouse prompt

A compact request with large hidden state

Design a low-power **64-bit RISC-V-based** **compute subsystem**
for an **XRBench-class real-time mobile XR workload**,
realize it as a **vector-capable CPU, accelerator, or SoC block**,
under a **3 W TDP target in a 3 nm-class LP mobile process**,
and return a **design-space report with evidence and rejected alternatives**.

Architecture obligations

What the prompt immediately forces an architect to define

XRBench workload
multi-model streams, deadlines

ISA and ABI contract
RISC-V baseline, extensions, compatibility

CPU / accelerator / SoC block
partitioning, vector support, interfaces

Memory and data movement
locality, hierarchy, interconnect energy

3 W power envelope
activity, thermal, energy/QoE tradeoff

Compiler/runtime stack
codegen, libraries, deployment

3 nm-class physical constraints
process, timing, leakage, signoff

Reliability and verification
correctness, corner cases, rejection

Design-loop requirements

One design turn: represent, act, gather evidence, reject, decide, revise

Intent / task → Representation / model → Tool / environment

feedback updates model

Compound methods → Evidence → Reject → Decide

decision revises the next loop turn

Boundary condition

Prompt-to-loop design with evidence, rejection, and judgment,
not prompt-to-chip automation.